

B.Sc (HONS.) IN CSE FIRST YEAR SECOND SEMESTER EXAMINATION, 2023
DISCRETE MATHEMATICS

[According to the New Syllabus]

Subject Code : 510223

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|--|-------|
| 1. (a) Define proposition, tautology, contradiction, contingency with truth table. | 6 |
| (b) Show that $\neg(p \vee (\neg p \wedge q))$ are $\neg p \wedge \neg q$ are logically equivalent. | 5 |
| (c) Let p and q be the propositions
p : You drive over 65 miles per hour
q : You get a speeding ticket
Write these propositions using p and q and logical connectives :
(i) You do not drive over 65 miles per hour.
(ii) You will get a speeding ticket if you drive over 65 miles per hour. <i>$p \rightarrow q$</i>
(iii) If you do not drive over 65 miles per hour, then you will not get a speeding ticket. <i>$\neg p \rightarrow \neg q$</i> | 5 |
| (d) Find bitwise OR, bitwise AND and bitwise XOR of bit strings 1111 0000, 1010 1010. | 4 |
| 2. (a) Prove that $\overline{A \cup B} = \bar{A} \cap \bar{B}$. | 5 |
| (b) Define disjoint set with example. Suppose that $A = \{1, 2, 4, 5, 6\}$, $B = \{2, 4, 6\}$. Find $B - A$. | 5 |
| (c) Use mathematical induction to show that
$a + ar + ar^2 + \dots + ar^n = \frac{ar^{n+1} - a}{r - 1} \text{ when } r \neq 1.$ | 5 |
| (d) Find the Fibonacci numbers f_2, f_3, f_4 and f_5 . | 5 |

[Please turn over]

Marks

3. (a) What is relation and binary relation with example.

4

- (b) Consider the following relation on $\{1, 2, 3, 4\}$

6

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$R_6 = \{(3, 4)\}$$

Which of these relations are reflexive, symmetric, antisymmetric and transitive?

- (c) Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find R^4 .

5

- (d) Suppose that the relations R_1 and R_2 on a set A are represented by

5

$$\text{the matrices } M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

What are matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$.

4. (a) Define Graph, Multigraph and Pseudograph. Construct a precedence graph for the following program :

6

$$S_1 : X := 0$$

$$S_2 : X := X + 1$$

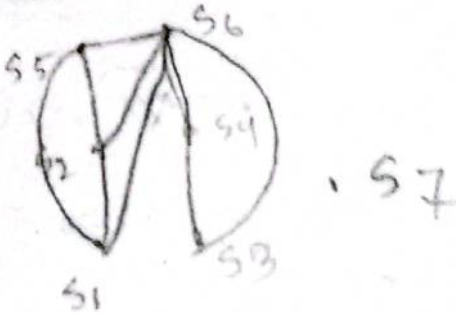
$$S_3 : Y := 2$$

$$S_4 : Z := Y$$

$$S_5 : X := X + 2$$

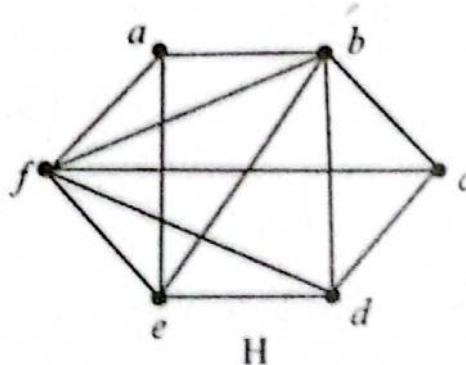
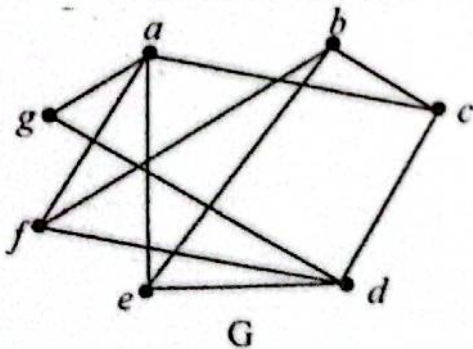
$$S_6 : Y := X + Z$$

$$S_7 : Z := 4$$



- (b) What is bipartite graph? Are the graphs G and H displayed in the following figure bipartite?

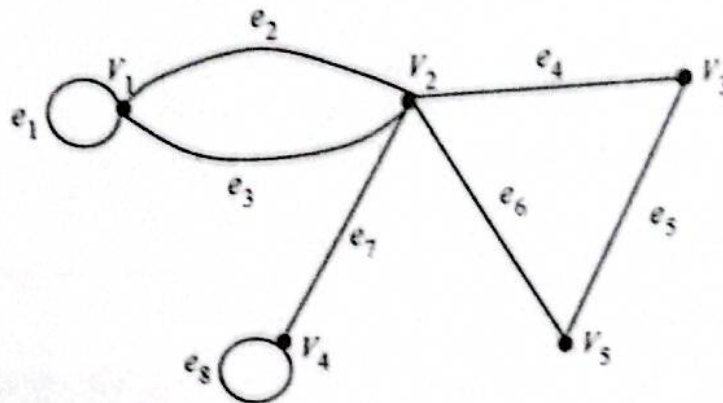
5



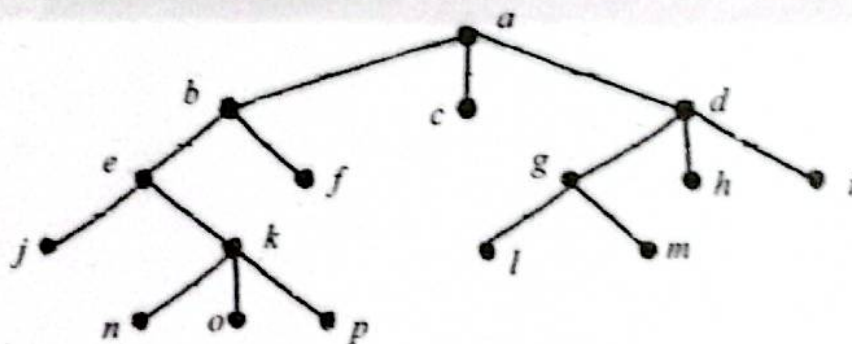


Marks

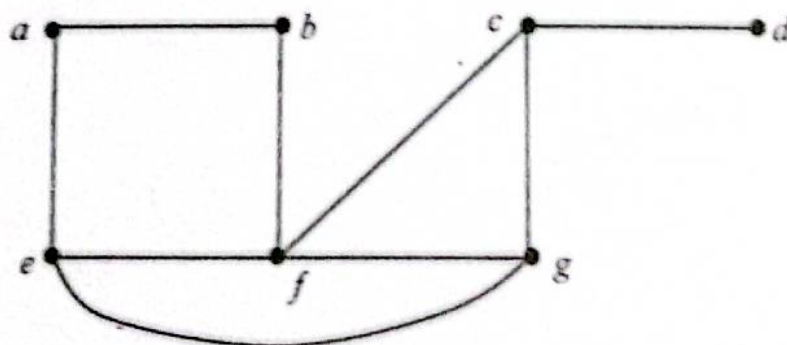
- (c) Represent the pseudograph shown in the following figure using an incidence matrix : 4



- (d) Define planar graph and chromatic number of a graph. What is the chromatic number of K_n ? 5
5. (a) Define rooted tree. Write down the properties of trees. 5
- (b) In which order does a preorder traversal visit the vertices of the following ordered rooted tree : 6



- (c) What is the value of the postfix expression $7\ 2\ 3\ * - 4\ \uparrow\ 9\ 3\ /\ +$? 5
- (d) Define spanning tree. Produce three different spanning trees for the simple graph shown in the following figure : 4



[Please turn over

- | | Marks |
|---|-------|
| 6. (a) Define Boolean expression and Boolean function. Find the values of the Boolean function represented by $F(x, y, z) = xy + \bar{z}$. | 5 |
| (b) What is duality? Find the duals of $x(y + 0)$ and $\bar{x} . 1 + (\bar{y} + z)$. | 3 |
| (c) Construct a circuit for a full adder using two half adders. | 6 |
| (d) Use Karnaugh maps to simplify the following sum-of-product expansions : | 6 |
| (i) $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$ | |
| (ii) $x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z$ | |

B.Sc (HONS.) IN CSE, FIRST YEAR, SECOND SEMESTER EXAMINATION, 2023
LINEAR ALGEBRA

[According to the New Syllabus]

Subject Code : 510225

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

Marks

1. (a) Define degenerate and nondegenerate linear equations. Solve the following system of linear equation : 3+5
=8

$$x_1 - x_2 + 2x_3 = 5$$

$$2x_1 + x_2 - x_3 = 2$$

$$2x_1 - x_2 - x_3 = 4$$

$$x_1 + 3x_2 + 2x_3 = 1$$

- (b) Determine the values of λ such that the following system of linear equations in x, y, z has (i) a unique solution (ii) no solution (iii) more than one solution : 6

$$x + y + \lambda z = 1$$

$$x + \lambda y + z = \lambda$$

$$\lambda x + y + z = \lambda^2$$

(c) Show that,

$$\begin{vmatrix} -1 & b & c & d \\ a & -1 & c & d \\ a & b & -1 & d \\ a & b & c & -1 \end{vmatrix} \begin{matrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - R_1 \\ R_4' = R_4 - R_1 \end{matrix}$$

$$= (a+1)(b+1)(c+1)(d+1) \left[1 - \frac{a}{a+1} - \frac{b}{b+1} - \frac{c}{c+1} - \frac{d}{d+1} \right].$$

[Please turn over]

a+1-q

2. (a) Define: Upper and Lower triangular matrices, Orthogonal matrix, Idempotent matrix. Marks 2+3 = 6

(b) Prove that the matrix $A = \frac{1}{6} \begin{bmatrix} 3 & 3 & 3 & 3 \\ 3 & -5 & 1 & 1 \\ 3 & 1 & 1 & -5 \\ 3 & 1 & -5 & 1 \end{bmatrix}$ is orthogonal. 6

(c) Solve the following linear equations with the help of matrices : 8

$$3x + 5y - 7z = 13$$

$$4x + y - 12z = 6$$

$$2x + 9y - 3z = 20$$

$$A^{-1} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

3. (a) Define distance and norm in $\mathbb{R}^n$. Consider the vectors 6

$\bar{x} = (1, -2, 3, -1)$, $\bar{y} = (2, 1, -5, 0)$ and $\bar{z} = (5, -3, -1, 1)$. Find

(i) $2\bar{x} + 3\bar{y} - 5\bar{z}$

(ii) $\bar{x} \cdot \bar{y}$

(iii) $\|\bar{x} - \bar{y}\|$

(iv) $d(\bar{x}, \bar{z})$.

$$4 \rightarrow 4 \cdot 8 = 4$$

$$2\pi_2 + \pi_2 = -1$$

$$\begin{matrix} -2 & 8 & -4 & 3 \\ 2 & 2 & 3 & 3 \end{matrix}$$

(b) If $\bar{u}, \bar{v} \in \mathbb{R}^n$, then show that,

$$\|\bar{u} + \bar{v}\|^2 + \|\bar{u} - \bar{v}\|^2 = 2(\|\bar{u}\|^2 + \|\bar{v}\|^2) \text{ and give its geometric interpretation.}$$

(c) If $\bar{u} = (1 + i, 2 - i)$ and $\bar{v} = (3 - 2i, 4 + i)$ then prove that, $\pi_2 - 2\pi_1$

$$\bar{u} \cdot \bar{v} = \overline{\bar{v} \cdot \bar{u}} \text{ and } d(\bar{u}, \bar{v}) = d(\bar{v}, \bar{u}).$$

$$2\pi_3 + \pi_2$$

4. (a) Define norm of a vector. State and prove Cauchy-Schwarz theorem. 2+6 = 8

(b) Determine the value of α and β if x is orthogonal to both y and z , 6
where $x = (1, \alpha, 2, -1)$, $y = (2, 1, 3, \beta)$ and $z = (\beta, 1, -4, 2)$.

(c) For any vector $u, v, w \in C^n$ then prove that:

(i) $(u + v) \cdot w = u \cdot w + v \cdot w$

(ii) $w \cdot (u + v) = w \cdot u + w \cdot v$

$$\begin{matrix} 0 & 5 & -5 \\ 0 & 7 & -1 \end{matrix}$$

$$\begin{matrix} 4 \rightarrow 2 & -2 & 7 & -1 \\ & 2 & -1 & 3 \\ & -2 & -4 & -8 \end{matrix}$$

5. (a) Define linear transformation. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be given by $T(x, y, z) = (x, y, 0) \forall x, y, z \in \mathbb{R}$. Show that T is a linear transformation. 2+5
=7

- (b) Define rank and nullity of a linear transformation. If $T : u \longrightarrow v$ is a linear transformation then show that rank of T + nullity of $T = \dim u$. 2+6
=8

- (c) $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear operator defined by 5

$$T(x, y, z) = (x + 2y - z, 2x - y + 4z, 4x + 3y - 2z).$$

Find a basis and the dimension of the (i) Im T of T ,
(ii) Ker T of T .

6. (a) Define : Inverse matrix, Echelon matrix, Spectrum. 2+3
=6

- (b) By using row canonical form find A^{-1} and hence or otherwise solve the following equation: 6

$$A \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \\ 4 \\ 6 \end{pmatrix} \text{ when } A = \begin{pmatrix} 0 & -1 & -2 & -3 \\ 1 & 1 & 4 & 4 \\ 1 & 3 & 7 & 9 \\ 1 & -2 & -4 & -6 \end{pmatrix}$$

- (c) Find the all Eigen values and the corresponding eigen vectors 8

for the matrix : $A = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{pmatrix}$.

Is the matrix diagonalizable? Also find an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix.

$$\begin{matrix} -2 & 8 & -4 \\ 2 & -1 & 3 \\ 0 & 7 & -1 \end{matrix}$$

$$\frac{a+1-a}{a+1}$$

$$= \frac{a+1}{a+1} - \frac{a}{a+1}$$

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**B.Sc. (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER
EXAMINATION, 2023**

DIGITAL SYSTEMS DESIGN

[According to the New Syllabus]

Subject Code: 510221

Examination Code : 5612

Time—3 hours

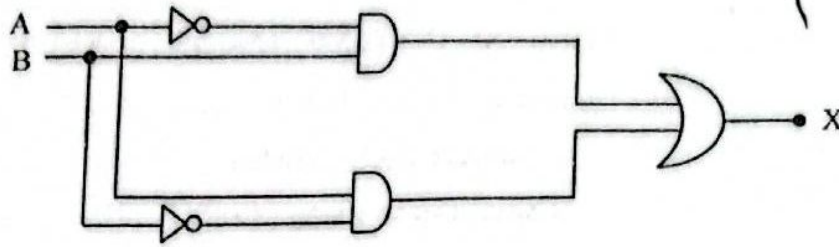
Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

- | | Marks |
|---|-------|
| 1. (a) What is analog and digital system? | 2 |
| (b) Write down the advantages of digital techniques. | 4 |
| (c) Convert the following numbers into given base numbers: | 6 |
| (i) $(562.65)_{10} = (?)_2 = (?)_8$ | |
| (ii) $(BCD)_{16} = (?)_8$ | |
| (d) Define the following terms with example : | 8 |
| (i) Hexadecimal Code | |
| (ii) BCD | |
| (iii) Unicode | |
| (iv) Excess 3 Code. | |
| 2. (a) Design a logic circuit with inputs P, Q and R. The output will be high whenever $P = 0$ or $Q = R = 1$. | 6 |
| (b) List the applications of Encoder and Decoder. | 4 |
| (c) What is Encoder? Design an octal to binary encoder. | 5 |
| (d) What do you mean by shift register? Mention different types of shift register. | 1+4=5 |

[Please turn over]

3. (a) Define min term and max term. (Prove that $X+X=X$ and $X \cdot X=X$). Marks
2+4=6
 (b) Determine the output expression for the following circuit: 4



- (c) What is karnaugh map? Minimize the following logic functions and realize using NOR gates: 1+5=6

$$F(A, B, C, D) = \pi M(1, 2, 3, 8, 9, 10, 11, 14). d(7, 15)$$

- (d) Explain with proper diagram of a clocked S—R flip-flop. 4

4. (a) What is binary parallel adder? Describe a 4-bit parallel adder with neat sketch. 1+5=6

- (b) Draw the logic circuit of latch and truth table using NAND gates. 4

- (c) Reduce the following function using K-map: 5

$$F = \overline{C}(\overline{A}\overline{B}\overline{D} + D) + A\overline{B}C + \overline{D}$$

- (d) Explain parity generator and checker with suitable diagram. 5

5. (a) What is subtractor? Describe full subtractor circuit with logic diagram and truth table. 7

- (b) Define ripple counter. Design a 4-bit ripple counter and sketch its transitional diagram. 8

- (c) Design a J-K flip-flop and describe its functions using truth table. 5

6. (a) Write down the function of transducer. 3

- (b) What is DAC? Explain the operation of DAC with proper diagram. 1+6=7

- (c) A certain 12-bit BCD digital to analog converter has a full scale output of 9.99V. Determine— 6

(i) Percentage resolution (ii) The converter's step size.

- (d) What are the advantages of R/2R ladder DAC over binary weighted registers. 4